

Kennect.ai

APM CASE STUDY



PROBLEM STATEMENT

Non-technical business users (incentive admins, MIS teams) regularly work with large and complex datasets to calculate incentives and performance reports.

However, existing data transformation tools are built for technical users, making tasks like joins, filters, and aggregations difficult, error-prone, and intimidating.

As a result, users rely on manual workarounds, repeatedly validate outputs, and risk incorrect business decisions.

There is a need for an intuitive, low-cognitive-load solution that enables non-technical users to safely transform data with confidence and accuracy.



PRATIK PATIL

IIT ROPAR

WHO ARE THE USERS



PAIN POINTS

A small white robot with a blue visor and a blue light on its chest. Next to it is a light green speech bubble containing the text 'Technical barriers for non-technical users?'.

Technical barriers for non-technical users?

Incentive admins and MIS staff often face complex data tasks (joins, filters, aggregations) using tools designed for analysts. Unfamiliar terminology and interfaces cause frustration and frequent errors

A small white robot with a blue visor and a blue light on its chest. Next to it is a light grey speech bubble containing the text 'Time-consuming manual work?'.

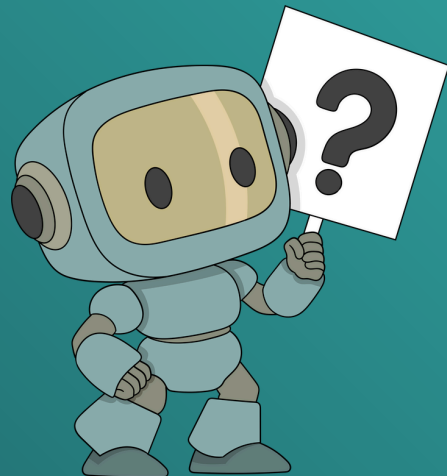
Time-consuming manual work?

Many users resort to manual spreadsheets or developer-led queries, leading to bottlenecks. Industry research shows business users spend up to 80% of their time on data preparation rather than analysis

A small white robot with a blue visor and a blue light on its chest. Next to it is a dark teal speech bubble containing the text 'High cognitive load?'.

High cognitive load?

Overwhelming options and lack of guidance can lead to “analysis paralysis.” Users struggle to formulate queries and reconcile data context. This drives error-prone work and delays in getting insights.



Stepwise pipeline with previews

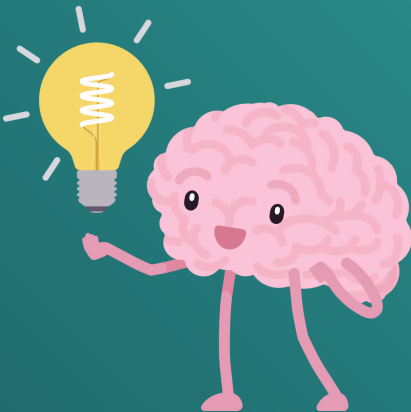
No-code visual transform builder

- Provide a form-based, drag-and-drop interface where users combine operations (filter, join, group-by, aggregate) as modular blocks. Each operation appears as an intuitive “building block” (like Lego pieces) that users stack to define the data workflow.

Familiar language & progressive disclosure

Use everyday business terms (e.g. “Filter by Date” rather than SQL) and reveal only essential options upfront. Advanced settings (custom formulas, complex joins) are hidden initially and can be accessed as needed. Aligning the UI to users’ mental models (visual metaphors, business contexts) accelerates onboarding (up to 60% faster training)

Minimal cognitive load design



Show transformations in sequence on a canvas (data → step1 → step2 → result). Include real-time previews after each step so users can verify outcomes immediately. This “what-you-see-is-what-you-get” approach removes guesswork and prevents errors

- **Keep the interface clean and uncluttered.** Use clear icons and consistent patterns (e.g. standard filter/merge symbols) so users build on familiar patterns. Employ smart defaults and contextual tooltips to offload decision-making.

USERS & PAIN POINTS

PROPOSED SOLUTION

PRIORITIZATION

RESEARCH & INSPIRATION

SUCCESS MATRICS

Form-based blocks: Filter, Join (basic), Group-by, Aggregate

Drag-and-drop canvas + dataset cards

- Why: Covers 80% of incentive workflows.
- Acceptance: Each block exposes 3–4 simple form fields (column, operator, value / key pairs for join).
- Impact/Effort: High impact / Medium effort.

- Why: Core mental model; users need to see pipeline.
- Acceptance: Can add/remove/reorder blocks; save workflow.
- Impact/Effort: High impact / Low–Medium effort.

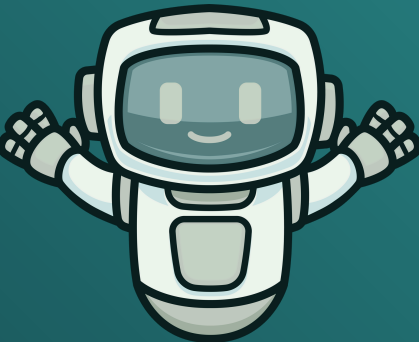
MVP NOW

Basic validation & safety checks (null-heavy warnings)

Sample-based live preview & row/column change badge

- Why: Prevents common mistakes that cause payout errors.
- Acceptance: Shows clear inline warnings and prevents final apply if critical mismatch.
- Impact/Effort: High impact / Medium effort.

- Why: Builds confidence and prevents errors.
- Acceptance: Shows sample records + “Rows before → after” + a short warning if row loss > X%.
- Impact/Effort: High impact / Low effort.



FUTURE (2-3 MONTHS)

Smart defaults & templates (common incentive recipes)

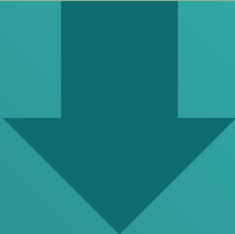
- Acceptance: “Monthly payouts” and “Region summary” templates available and editable.
- Impact/Effort: High impact / Low effort.

Advanced join modes + join diagnostics (preview unmatched keys, heatmap of key distributions)

- Impact/Effort: Medium impact / Medium effort.

Reusable workflows & sharing (collaboration)

- Impact/Effort: Medium impact / Low–Medium effort.

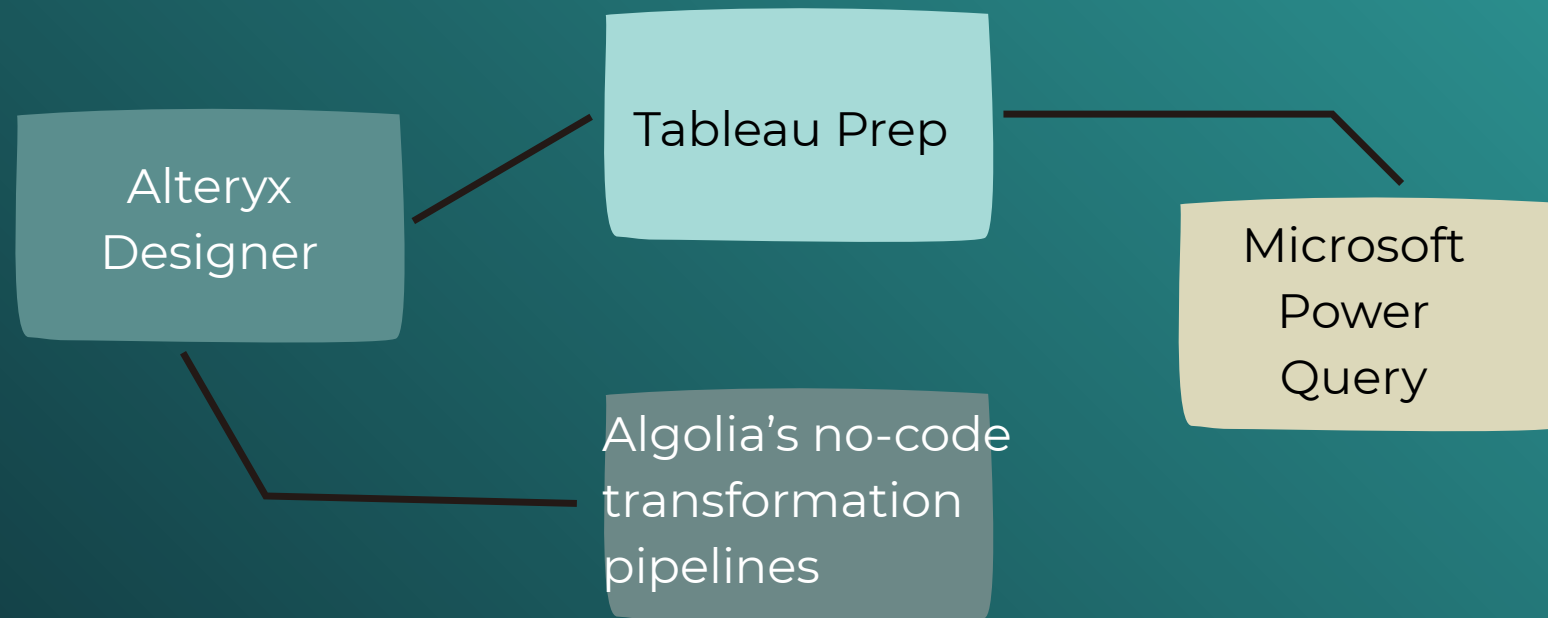


FUTURE (0.5-1 YEAR)

Natural-language helper (NLU “show me total sales by region”)

AI-assisted suggestions & anomaly detection (suggest join keys, flag suspicious outputs)

Multi-table joins, window functions, custom formulas editor for power users

UNDERSTANDING THE
CURRENT PROCESS

Industry-backed design inspiration

- Visual pipelines increase task success
No-code, visual workflows have been shown to improve task completion and reduce user errors compared to query-based tools.
- Drag-and-drop lowers learning curve
Workflow-based UIs enable faster onboarding by aligning with how users naturally think in steps and sequences.
- Preview-first interfaces reduce mistakes
Immediate feedback after each step prevents late-stage errors and rework.
- Composable blocks scale complexity safely
Breaking transformations into modular blocks allows users to build complex logic incrementally without cognitive overload.

- **Analyze industry examples:** Products like Alteryx Designer use a drag-and-drop workflow canvas that “empowers analysts” with a visual pipeline. Tableau Prep and Microsoft Power Query likewise offer direct, no-code data prep flows. The Algolia search platform’s no-code data transformations use stackable blocks to simplify operations. We can draw on these for inspiration.
- **Design patterns:** A clean flowchart-style canvas is key. Arrange transformations left-to-right (or top-down) to mimic a data pipeline. Use well-known icons (table, funnel, gears) for actions. Leverage a sidebar or toolbar with common actions (join, sort, filter icons) so users can quickly add steps.
- **Visual hierarchy and cues:** Highlight the current step in the workflow with a glow or border. Use breadcrumbs or a step counter at the top. Color-code block types (e.g. blue = filter, green = join, orange = output) for quick recognition.
- **Intuitive layout:** Present data sources as nodes or cards that flow into the transformation canvas. Allow dragging tables or fields onto join/filter modules. Employ clear typography and ample spacing to reduce clutter.

MATRICS TO LOOK	Adoption	Efficiency Gains	Data Quality & Trust	User Satisfaction	Business Impact
	- Track how many target users (admins, MIS) run transformations weekly / monthly - Goal: High adoption (≥80% of eligible users) - Track growth in total transformation count over time ➤ Segment adoption by user persona (admin vs MIS) ➤ Monitor drop-offs after first use	- Measure time saved per task (before vs after) - Aim for ~80% reduction in data preparation time - Compare completion time for common workflows (join + aggregate) - Monitor reduction in support tickets / IT dependency ➤ Measure number of manual Excel exports avoided	- Track error rates: - Failed transformations - Data mismatches - Incorrect joins - Fewer manual corrections required post-export - Higher confidence scores via user surveys ➤ Track warning resolution rate (how often users fix issues highlighted by the system)	- Use CSAT / NPS to gauge ease-of-use - Track qualitative feedback: “Intuitive” “Fast” “Safe to use” - Target consistently high satisfaction ratings ➤ Track onboarding satisfaction separately for first-time users	- Faster report & incentive payout turnaround time - Increased data-driven decision-making across teams - Report overall ROI in: - Hours saved - Reduced operational overhead - Improved KPI accuracy ➤ Link impact to downstream metrics (retention, revenue confidence)

Pitfalls & Mitigations			What More Can Be Done?		
Potential Pitfalls	How to Reduce Risk		Reusable Transformation Templates		
	• Mandatory preview & validation before final output		Pre-built workflows for common MIS and incentive reporting use cases.		
• Users misinterpreting transformation results	• Row-loss and key-mismatch hard warnings		AI-Assisted Guidance		
	• Sample-based execution for large data volumes		Suggested join keys, anomaly detection, and smart defaults.		
• Incorrect joins causing silent data loss	• Contextual tooltips explaining why changes occurred		Collaboration & Governance		
	• Easy undo / version history for recovery		Workflow sharing, approvals, and audit logs for sensitive reports.		
• Over-simplification limiting complex use cases			Enterprise Readiness		
			Role-based permissions, compliance logging, and data lineage tracking.		
• Performance lag with large datasets (2M+ records)					

WIREFRAMES

Annotated Wireframe: No-Code Data Transformation Builder

Design a feature that allows a non-technical user to apply dataset transformations sin a way:

Grid

Filter

Table

Table

Table

Table

Sales Transactions

1.2M rows | 45 columns

↓

Filter Block

Region = West

↓

Join Block

Sales.Employee_ID ::> Left Join

Employee.Employee_ID

43,000 rows unmatched

↓

Group By

Month, Region

Aggregate. SUM(Incentive)

920K rows

LIVE PREVIEW

Rows: 1,200,000 → 1,080,000

⚠️ 10% rows unmatched (Employee_ID)

Month	Region	Employee_ID	Sales
Jan	West	1055	\$5,200
Jan	West	1036	\$4,800
Jan	West	1094	\$7,600

Sample rows shown (100)

Month	Region	Employee_ID	Incentive
Jan	West	\$5,200	\$.300
Jan	West	\$4,800	\$.260
Jan	West	\$7,600	\$.450

View unmatched rows

Edit

↩️ ✕

Apply

1

Business-friendly blocks

Each block represents one business action using simple forms, not SQL

2

Stepwise visual pipeline

Users clearly see the order of transformations, reducing confusion and mistakes

3

Live previews & warnings

Instant feedback on row loss or mismatches before applying changes

4

Safe experimentation

Users can edit, reorder, or undo steps without breaking data